

Software Testing

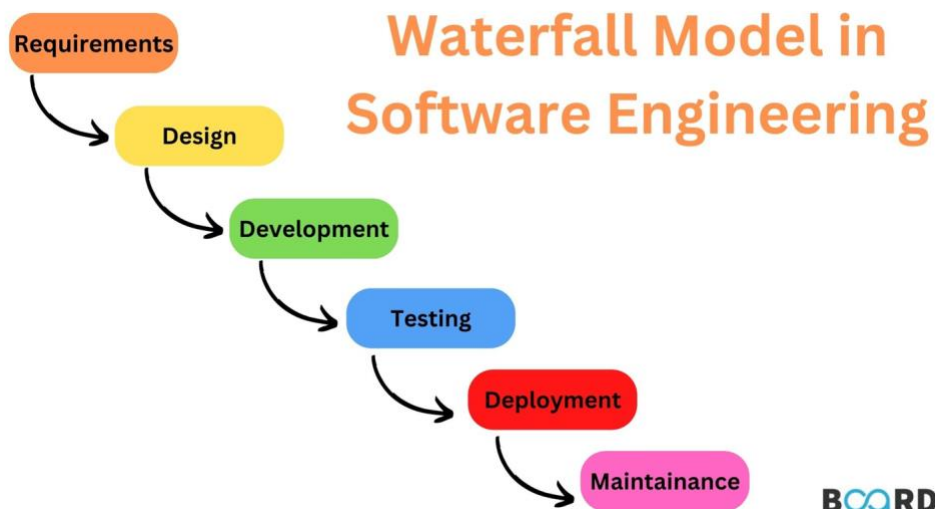
Software Testing is a method to check whether the actual software product matches expected requirements and to ensure that software product is Defect free.



It involves execution of software/system components using manual or automated tools to evaluate one or more properties of interest. The purpose of software testing is to identify errors, gaps or missing requirements in contrast to actual requirements.

WaterFall Model

The waterfall model is useful in situations where the project requirements are well-defined and the project goals are clear.



Agile Model

Agile Methodology meaning a practice that promotes **continuous iteration** of development and testing throughout the software development lifecycle of the project. In the Agile model in software testing, both development and testing activities are concurrent, unlike the Waterfall model.

The **Agile software development** methodology is one of the simplest and effective processes to turn a vision for a business need into software solutions

The agile software development emphasizes on four core values.

1. Individual and team interactions over processes and tools

No matter how well-researched your process and high-tech your tools are, it's the team you work with and the way you work together that determines success.

2. Working software over comprehensive documentation

Traditional product development processes often required extensive documentation before a single line of code was written. Under the agile philosophy, getting software in the hands of customers is the highest priority

3. Customer collaboration over contract negotiation

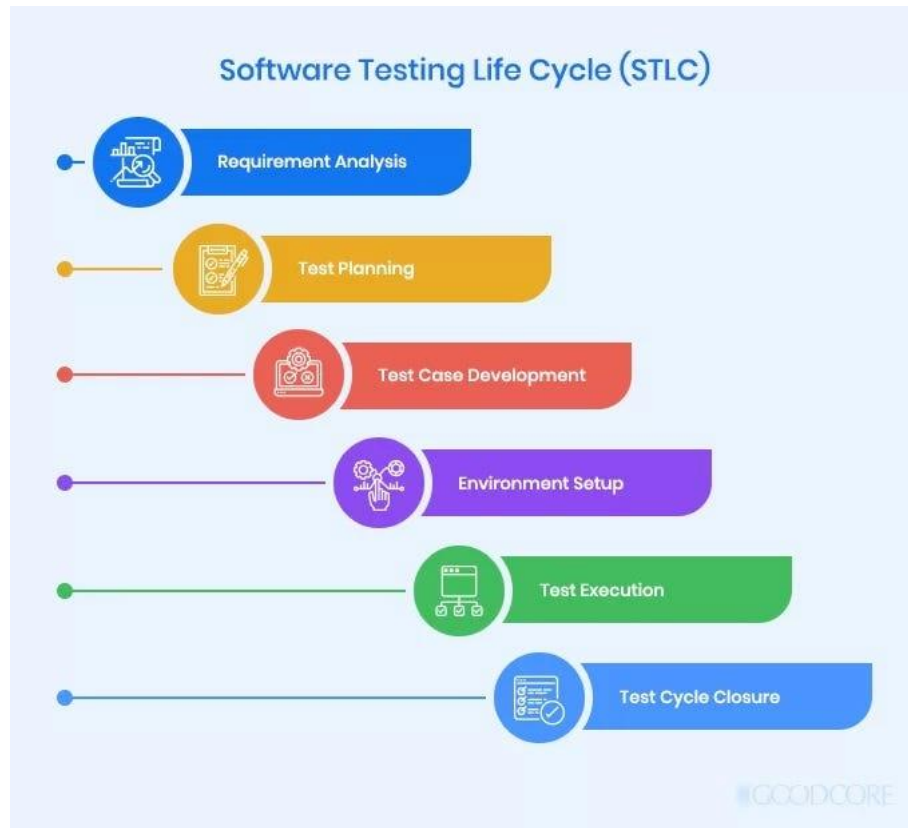
The agile philosophy highlights the importance of customer-centric product development practices over product-centric approaches. While contracts will always have their place in business, a list of the things you're offering your customer is no replacement for actually communicating with them about what their needs are and where their challenges are

4. Responding to change over following a plan

An important benefit of the agile methodology is that it encourages frequent reviewing and retooling of current plans based on new information that the team is continually gathering and analyzing.

STLC

Software Testing Life Cycle (STLC) is a sequence of specific activities conducted during the testing process to ensure software quality goals are met.



Functional Testing

Functional Testing is a type of software testing that validates the software system against the functional requirements/specifications. The purpose of Functional tests is to test each function of the software application, by providing appropriate input, verifying the output against the Functional requirements.

SMOKE TESTING

SANITY TESTING

REGRESSION TESTING

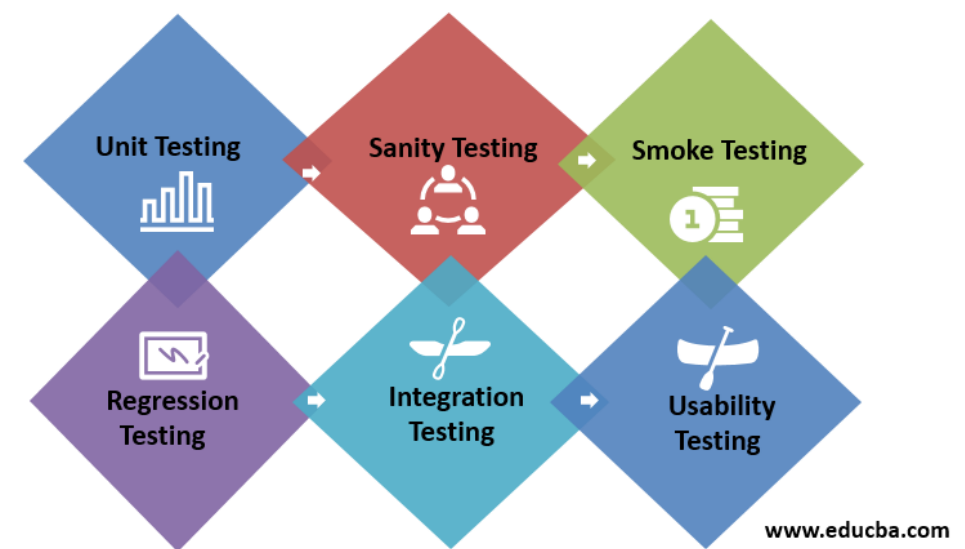
RETESTING

UNIT TESTING

INTEGRATION TESTING

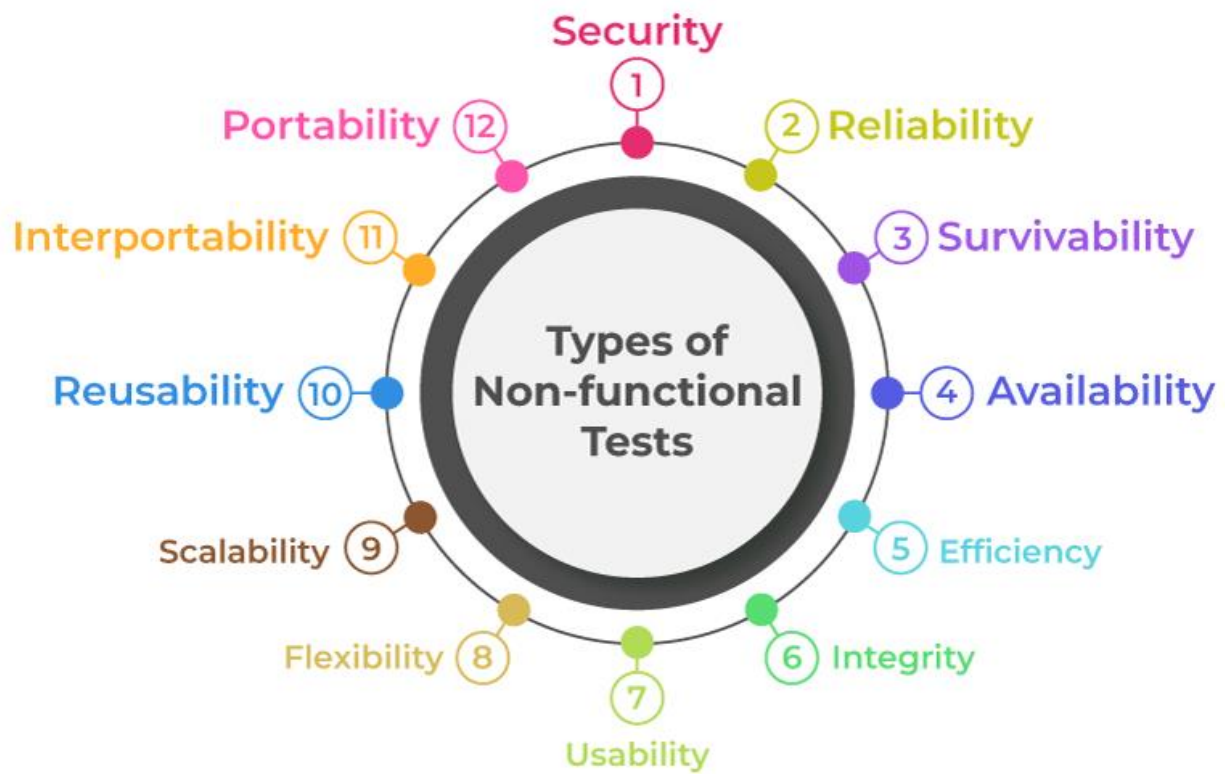
SYSTEM TESTING

Types of Functional Testing



Non Functional

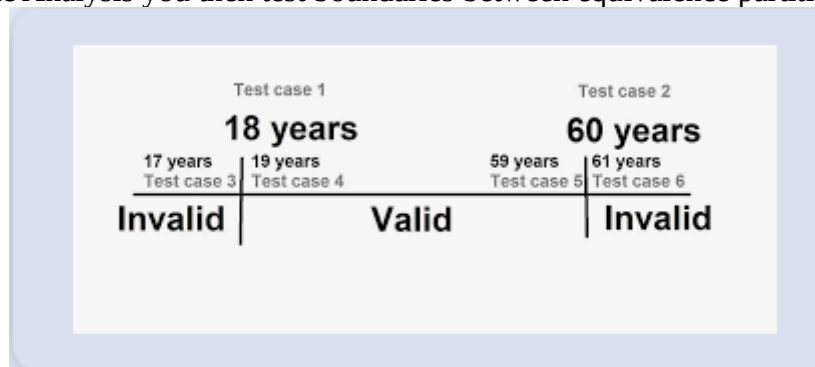
Non-Functional Testing is defined as a type of Software testing to check non-functional aspects (performance, usability, reliability, etc) of a software application. It is designed to test the readiness



of a system as per nonfunctional parameters which are never addressed by functional testing.

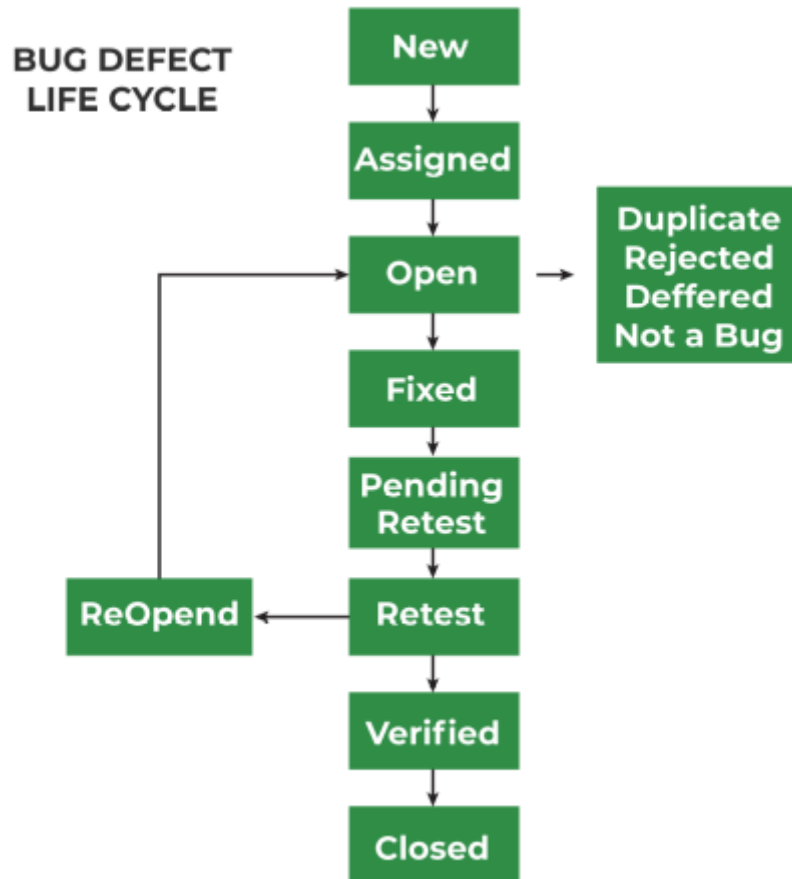
BVA

In Boundary Value Analysis you then test boundaries between equivalence partitions



Defect Life Cycle

Defect Life Cycle or Bug Life Cycle in software testing is the specific set of states that defect or bug goes through in its entire life. The purpose of Defect life cycle is to easily coordinate and communicate current status of defect which changes to various assignees and make the defect fixing process systematic and efficient



Priority is the order in which the developer should resolve a defect whereas **Severity is the degree of impact that a defect has on the operation of the product.**

Priority is categorized into three types: **low, medium and high** whereas Severity is categorized into five types: **critical, major, moderate, minor and cosmetic.**

Few very important scenarios related to the severity and priority which are asked during the interview:

High Priority & High Severity: An error which occurs on the basic functionality of the application and will not allow the user to use the system. (Eg. A site maintaining the student details, on saving record if it, doesn't allow to save the record then this is high priority and high severity bug.

High Priority & Low Severity: The spelling mistakes that happens on the cover page or heading or title of an application.

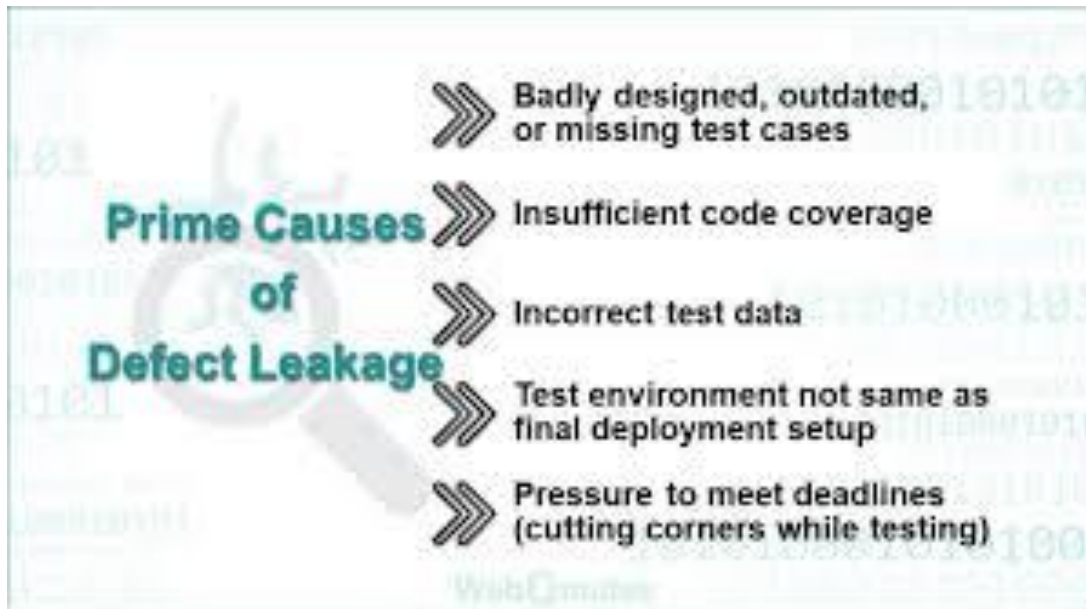
High Severity & Low Priority: An error which occurs on the functionality of the application (for which there is no workaround) and will not allow the user to use the system but on click of link which is rarely used by the end user.

Low Priority and Low Severity: Any cosmetic or spelling issues which is within a paragraph or in the report (Not on cover page, heading, title).

Defect Leakage

Defect leakage measures the percentage of problems that leak from one testing stage to the next, demonstrating the efficacy of testing.

Defect leakage in software testing to calculate the total number of undetected errors in software systems. This method also helps them validate their testing efforts by allowing them to determine the total number of faults present in a software program.



Testing

Environments -

Different Testing Environments

SIT

UAT

PRE PROD

PROD

Basic Concepts

Walkthrough,

Test Case Writing (**RTM, Test Scenarios, Test Cases**)

Timelines

Review

Execution

Defect Reporting

Summary Report

Sign Off

Daily Routine



9.30 – Login & start a day with Morning Hurdle Call (15 to 30 mins)

Plan of Action, New & or Pending Execution, Open Defects – Retesting.

10.00 – Start Execution as per PoA discussed in MHC.

11.00 – Stand Up Call with Client & other Stakeholders (15 to 30 Mins)

12.30 – On discussion with Client – Share Defect Tracker (First Half)

1.00 – Lunch

1.30 - Start Execution as per PoA discussed in MHC & Stand Up Call.

4.00 - Share Defect Tracker (Second Half)

7.00 – Connect with Client > Give Update of Testing Status > Share EOD Summary Email of Test Cases & Defect Tracker > EOD

THANK YOU